



PSOC™ 4 Inductive sensing (ISX) using PSOC™ 4000T and PSOC™ 4100T Plus

Customer training workshop (CTW)

October 2026



Table of contents

1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

Table of contents

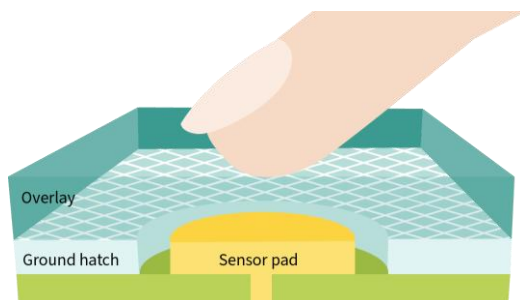
1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

Multi-sensing technologies with inductive and liquid sensing

Infineon PSOC™ 4 with Multi-sense technology

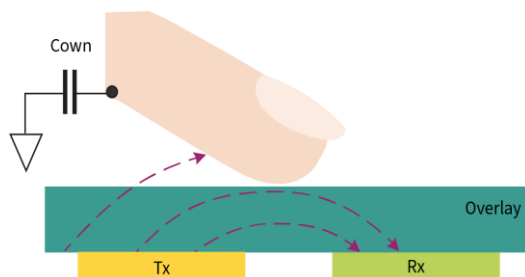
CAPSENSE™ – Capacitive sensing

Self-capacitive sensing



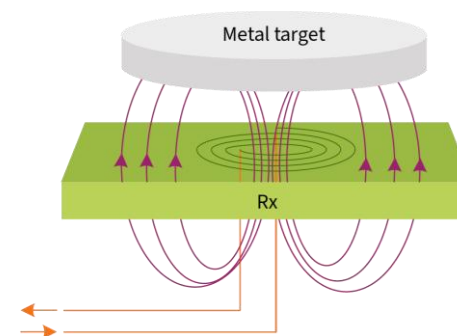
- **CSD** – Infineon's proprietary self-capacitance sensing method
- Measures capacitance of a single electrode
- User touch increases the capacitance of the electrode

Mutual capacitive sensing



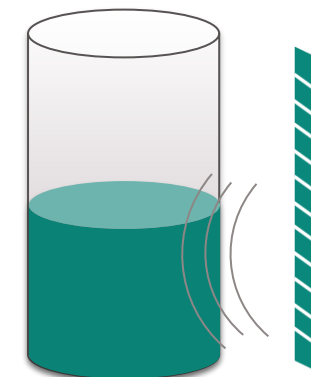
- **CSX** – Infineon's proprietary Mutual-capacitance sensing method
- Measures capacitance between two electrodes
- User touch decreases capacitance between electrodes

Inductive sensing



- **ISX** – Infineon's proprietary Inductive sensing method
- Measures inductance of sensor coil
- Deflection of metal decreases inductance of the electrode

Liquid sensing



- **LLS** – Infineon's proprietary liquid sensing method
- Detects presence and level of liquid without contacts
- Reliable sensing and rejects false signals such as foam mist and droplets

Use case overview and features

Infineon PSOC™ 4 with Multi-sense technology

CAPSENSE™ – Capacitive sensing

Self-capacitive sensing

- Proximity
- Buttons
- Sliders
- One-finger touchpad/ touchscreen
- Wear detection
- Humidity sensing
- Hover-touch

Mutual capacitive sensing

- Multi-finger matrix buttons
- Sliders
- Multi-finger touchpad/ trackpad
- Wear detection
- Humidity sensing
- Vibration and shock sensing

Inductive sensing

- Force touch
- Touch-over metal
- Underwater touch
- Proximity/Reed switch
- Wear and grip detection
- Metal and object detection
- Shaft and rotary encoders
- Vibration and shock sensing

Liquid sensing

- Liquid presence detection
- Liquid-level detection
- Foam detection / rejection

Best-in class sensing performance

- **Ultra-low-power** based on Always-on autonomous sensing operation
- **Superior signal-to-noise ratio** performance based on all-new ratio-metric and differential sensing
- **Multi-sense technology** in a single chip with an **innovative sensing architecture** for flexible, reliable, and easy design experience

Applications/use cases

– Touch-over-metal

- Integration into appliances with existing metal surfaces
- Coffee machines, microwave oven, refrigerator, washing machine, and dishwasher

– Underwater/waterproof touch

- Inherent immunity to water/liquids
- Underwater cameras, personal care products (shaver), Intercom, EV charging, and other systems that are exposed to harsh and wet environments, and washing machines

– Force touch buttons

- Detection of different force levels is possible, unlike CAPSENSE™
- Personal computer products (computers, tablets), Outdoor keypads (smart locks, security products), headphones wear detection, Power tools, buttons, and grip detection

– Proximity sensing and reed switch replacement

- Smart home (door locks, security systems), Home appliances (washing machines, ovens, refrigerators) and industrial applications

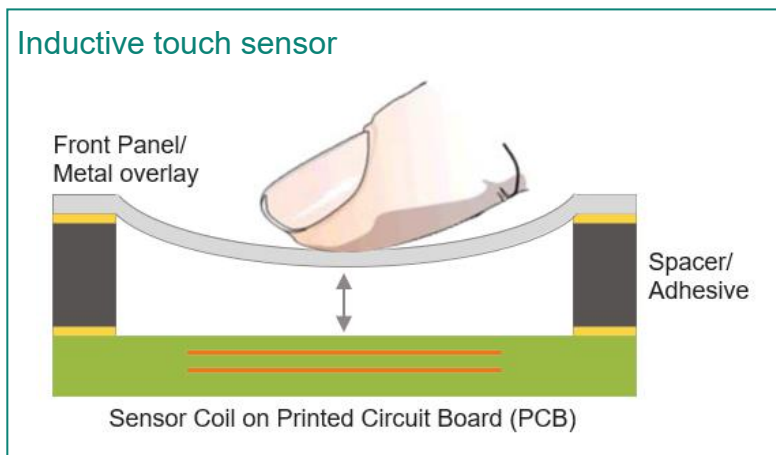
Table of contents

1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

Basic concept of inductive sensing

Introduction

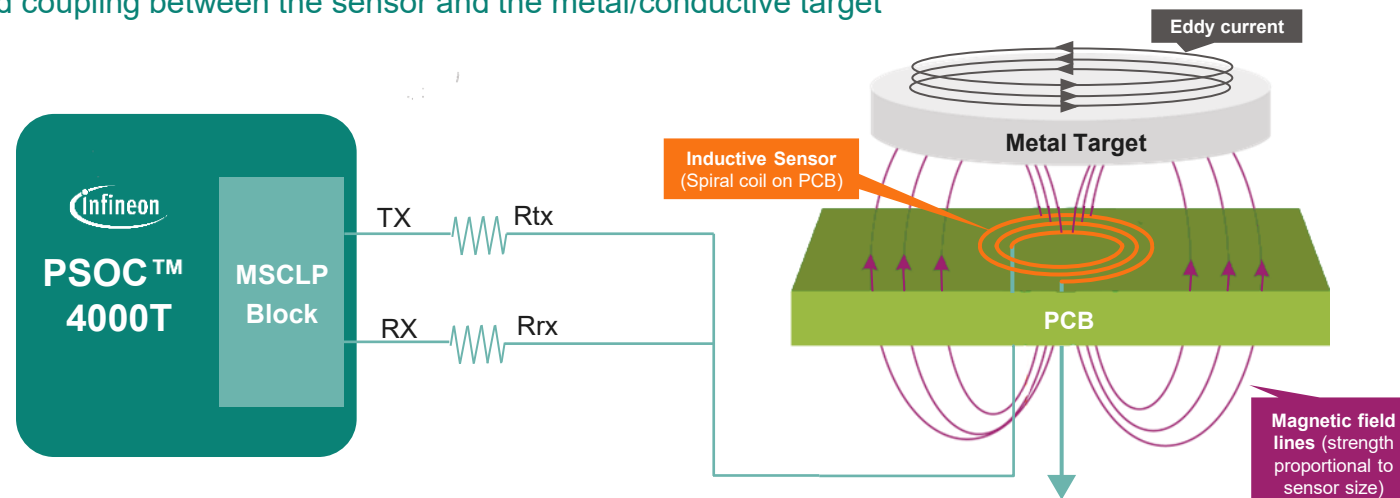
- Inductive sensing provides innovative, low-cost solution to detect the presence or movement of nearby metallic or conductive objects
- The technology enables robust, low-cost HMI with metallic overlays that seamlessly integrate into the product
- Inductive sensing addresses shortcomings of capacitive sensing and enables new and creative methods for touch HMI



Working principle

- Principle of **electromagnetic coupling between a sensor coil and the metal/highly conductive object** to be detected
- Electromagnetic energy is transferred from the sensor coil's electromagnetic field into the metal target. Transferred energy causes a circulating electrical current called **eddy current**
- The eddy current flowing in the metal target induces reverse electromagnetic field on the sensor coil, which results in a **reduction of the effective inductance of the sensor coil**

Field coupling between the sensor and the metal/conductive target

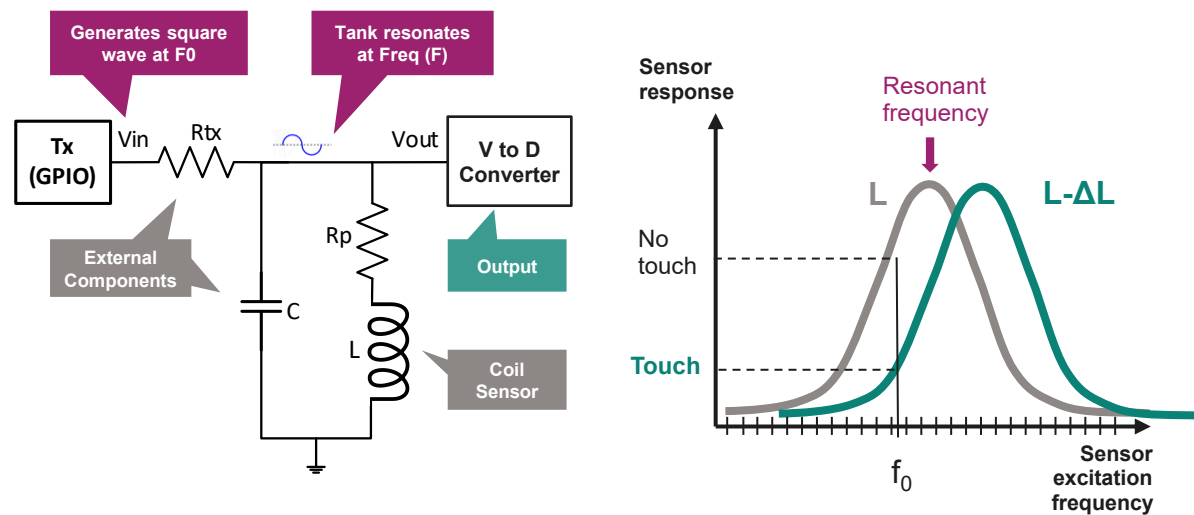


Inductive sensing method: L-C Tank oscillator

1. L-C Tank/resonance methods

- Can be realized with a voltage divider circuit or a frequency shift circuit
- Sensor coil inductor and external capacitor form an L-C tank oscillator. The voltages generated in the tank circuit are demodulated and converted as a voltage-mode signal
- Change in coil inductance shifts the L-C tank resonant frequency, resulting in a change in the voltage

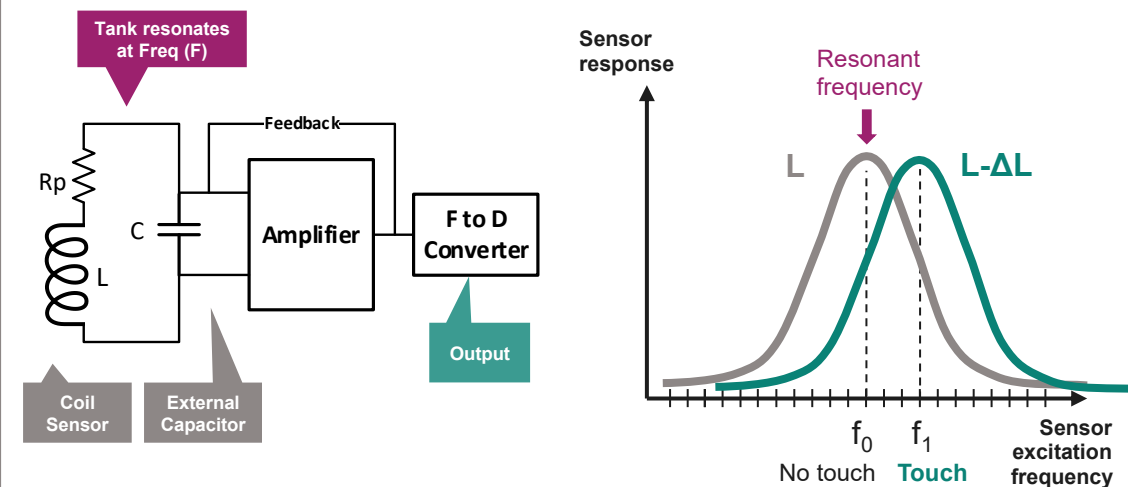
1a. L-C Tank voltage divider circuit



2. Fly-back method from Infineon

- Increases operation frequency range
- Improves noise immunity
- Reduces emissions
- Increases overall sensor design flexibility

1b. L-C Tank oscillator circuit

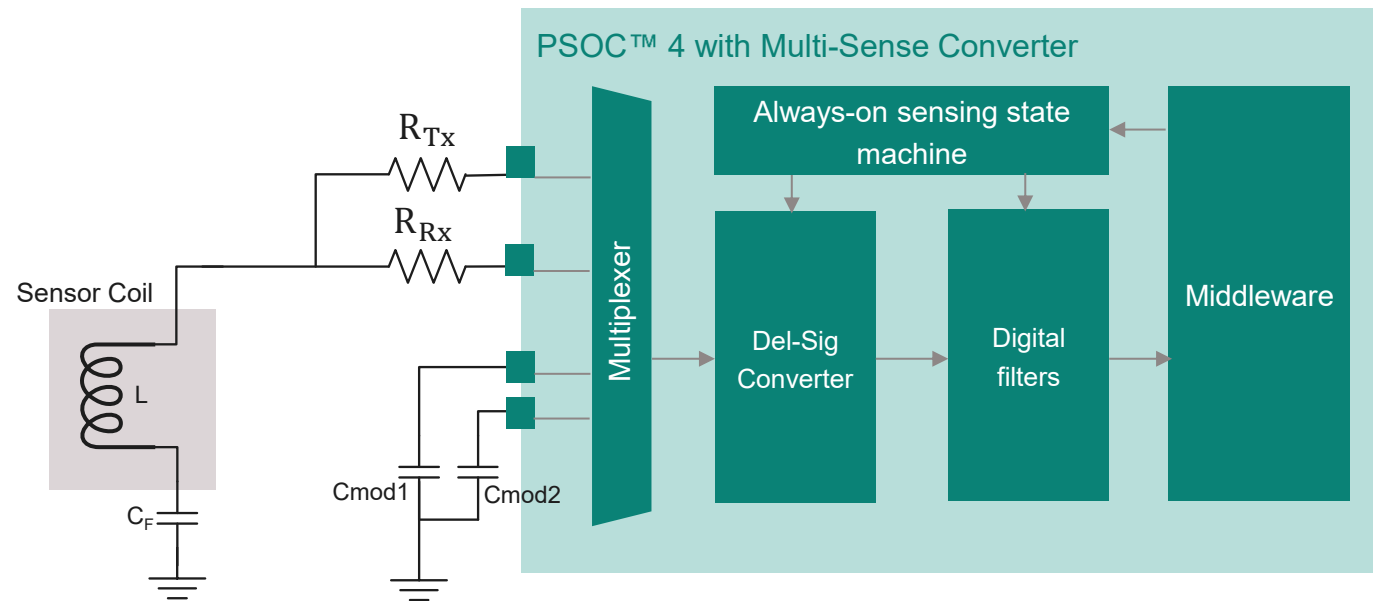


Inductive sensing architecture based on industry leading CAPSENSE™ engine

Superior performance based on differential and ratio-metric sensing architecture

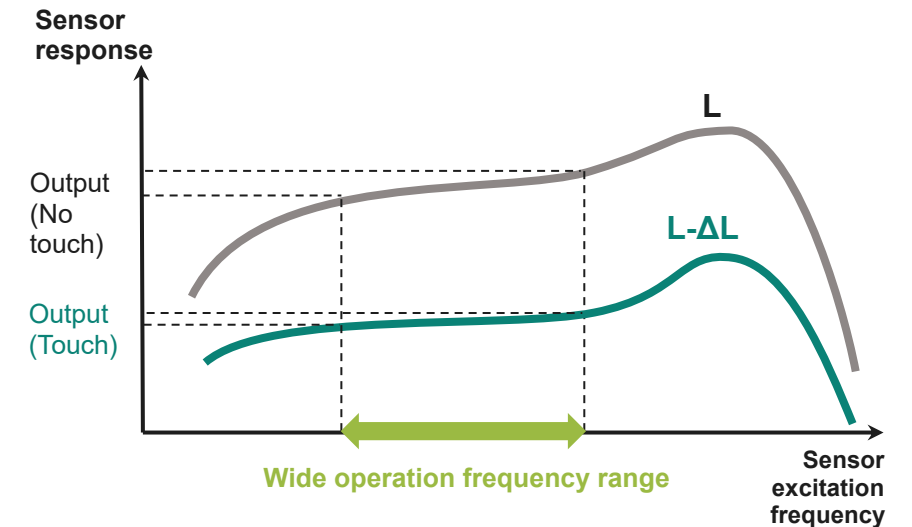
- Built on Infineon's **industry-leading CAPSENSE™ sensing engine** and years of learning in HMI sensing
- **Differential and ratio-metric sensing architecture** for state-of-the-art sensing performance, insensitive to clock and supply noises
- **Flexible analog frontend** enabling combination of Multi-Sense functions on any GPIOs, using a common pair of external Cmod capacitors

2. Infineon Fly-back method with Multi-Sense Converter and linear transfer function



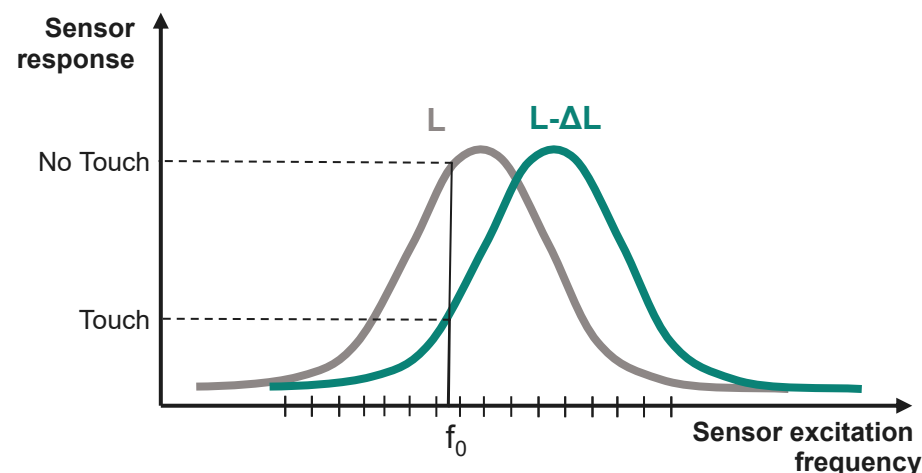
Note: C_f is floating capacitor; C_{ref} is internal reference cap

$$\text{Transfer function: } D_{out} = \sim \frac{1}{R_{Tx}R_{Rx}C_{ref}} \cdot L$$



Reliable performance across frequency and variations in fly-back

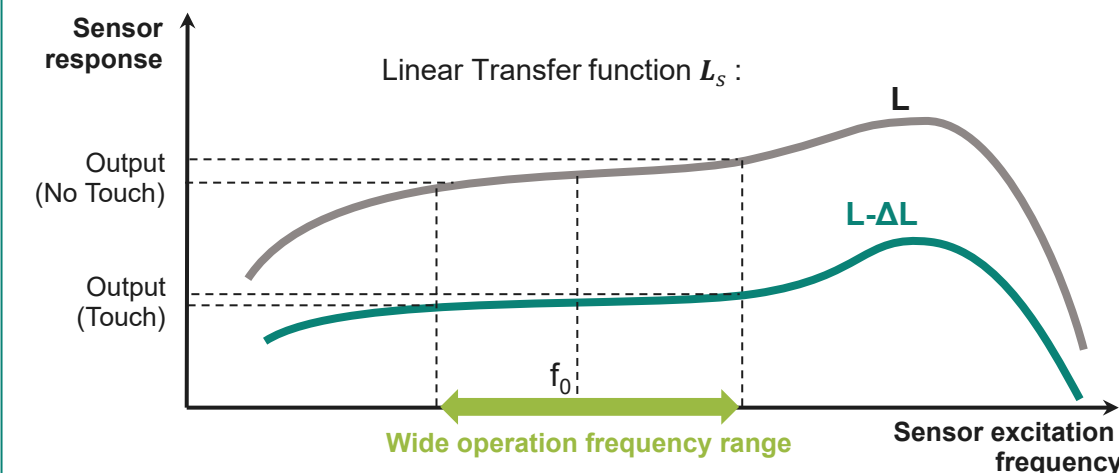
1. L-C Tank voltage divider circuit



Summary L-C Tank: Process variation lead to poor performance

- **Non-linear transfer characteristics** lead to complex design considerations
- Optimal performance is delivered only when the L-C tank is excited at the resonance frequency; if not, it leads to a **significant loss of signal**
- Critical for the sensor to be tuned at the resonance frequency for operation
- **Susceptible to external noise aggressors** and has no flexibility
- System performance is **prone to process variation or sensor design**, and external components

2. Infineon's Fly-back sensing method



Summary Fly-back: Consistent high performance

- Sensing circuit with **linear transfer characteristics**, easy to design
- **Wide flexibility** on both sensor inductance and sensor operating frequency without performance compromise
- **Faster learning curve**, easy to design and tune, wide operating freq. range
- **Exceptional flexibility** for system noise aggressor (EMC/EMI) mitigation incl. Spread-Spectrum Clock and frequency hopping
- **Consistent sensor output** regardless of sensor frequency or process variations

Infiniteon's Fly-back with significant benefits over L-C/resonance method



	Inductive sensing (Flyback method)	Inductive sensing (Voltage divider method*)	Inductive sensing (Oscillator method)
Wide operation frequency range	✓	✗	✗
Multiple operating frequencies	✓	✗	✗
Sensor design flexibility on PCB	✓	✗	✗
Noise immunity	✓	~ (no flexibility)	~ (no flexibility)
Noise emission	✓	~ (no flexibility)	~ (no flexibility)
Temperature stability	~	✗	✗
Liquid tolerant/proof	✓	✓	~ (larger inductors with large sensitivity)
No. of external components	~ (max. 3)	~ (0 to 5)	~(0 to 5)
External components manufacture tolerances stability	✓	✗	✗
Sensitivity	✓	✓	~ (poor for large inductors)
Linear response	✓	✗	✗

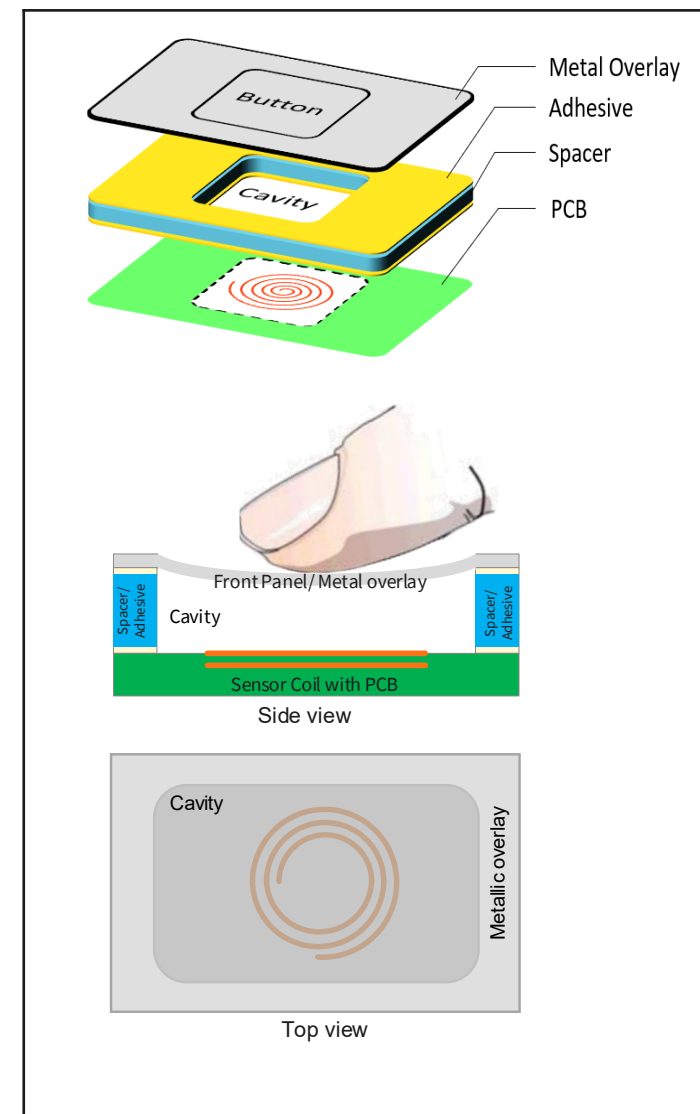
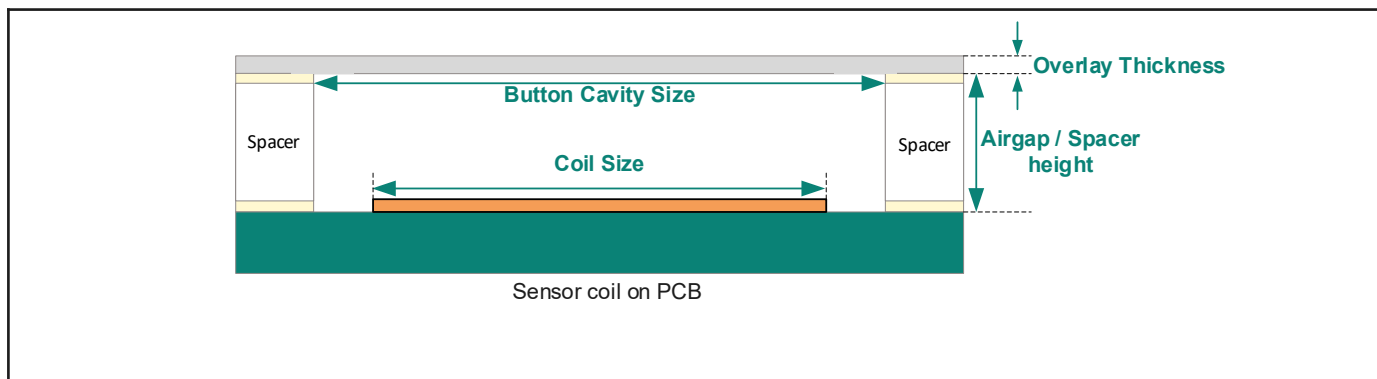
* Magsense v1

Table of contents

1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

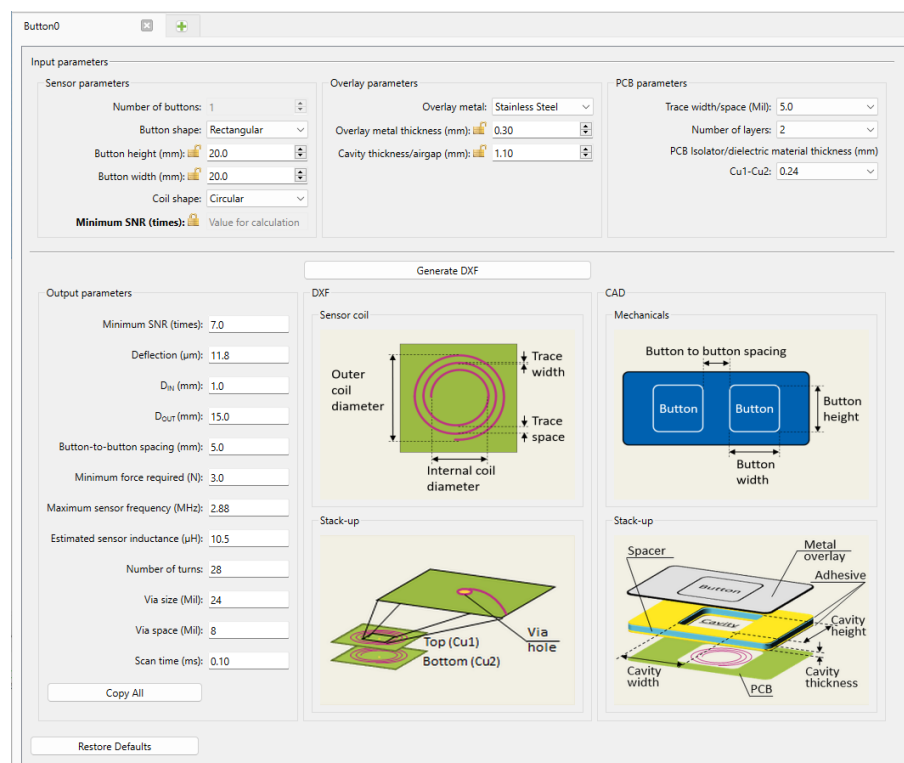
Sensor mechanicals

- For touch-over metal buttons, the mechanical stack-up plays a crucial role in sensor performance
- The following parameters are considered:
 - Button shape and size (the overlay on top)
 - Sensor shape and size (the actual coil on the PCB)
 - Overlay material and thickness(mm)
 - Spacer material
 - Spacer thickness/overlay-sensor-airgap (mm)
 - PCB specs(number of layers, trace width and so on)
- The mechanical construction should allow for good overlay deflection at the required force



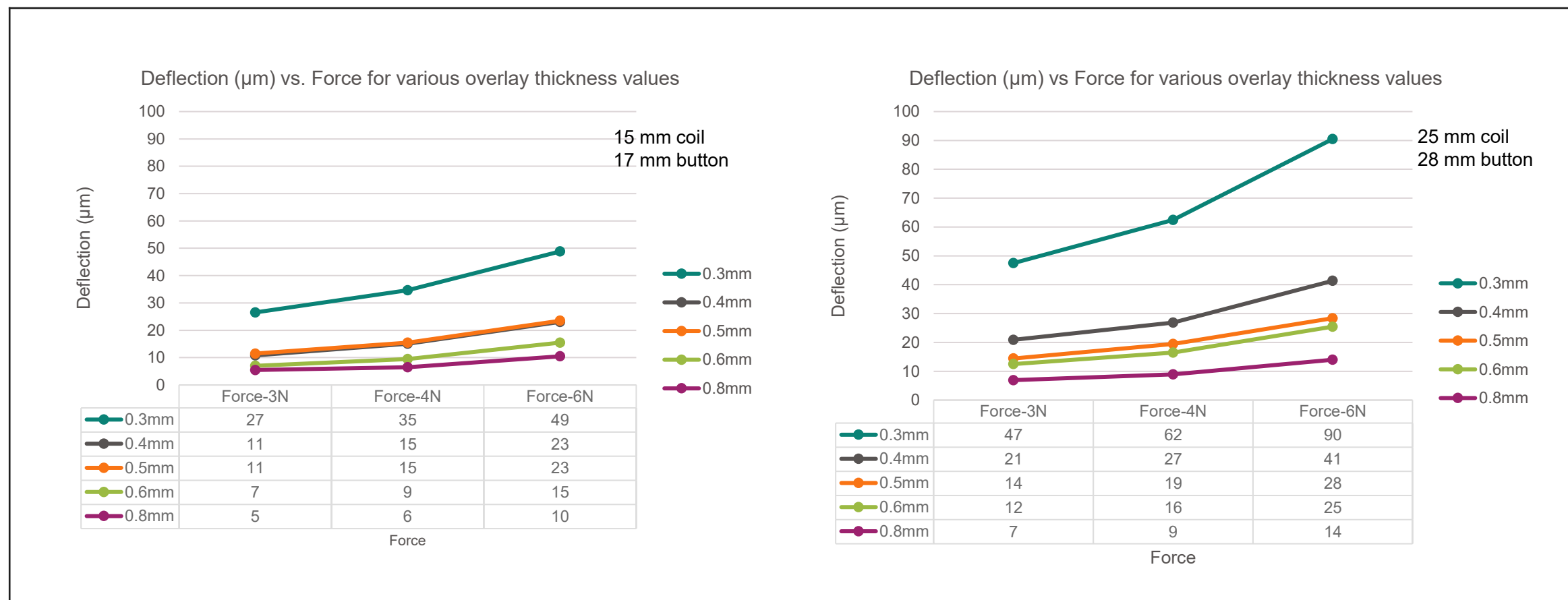
Sensor designer tool

- It takes in different design parameters as input and give a .dxf file which can then be integrated in the PCB/product design
- This tool can be launched directly from ModusToolbox™
- It will help customers estimate the inductance, SNR and so on to help arrive on a sensor design faster with some confidence



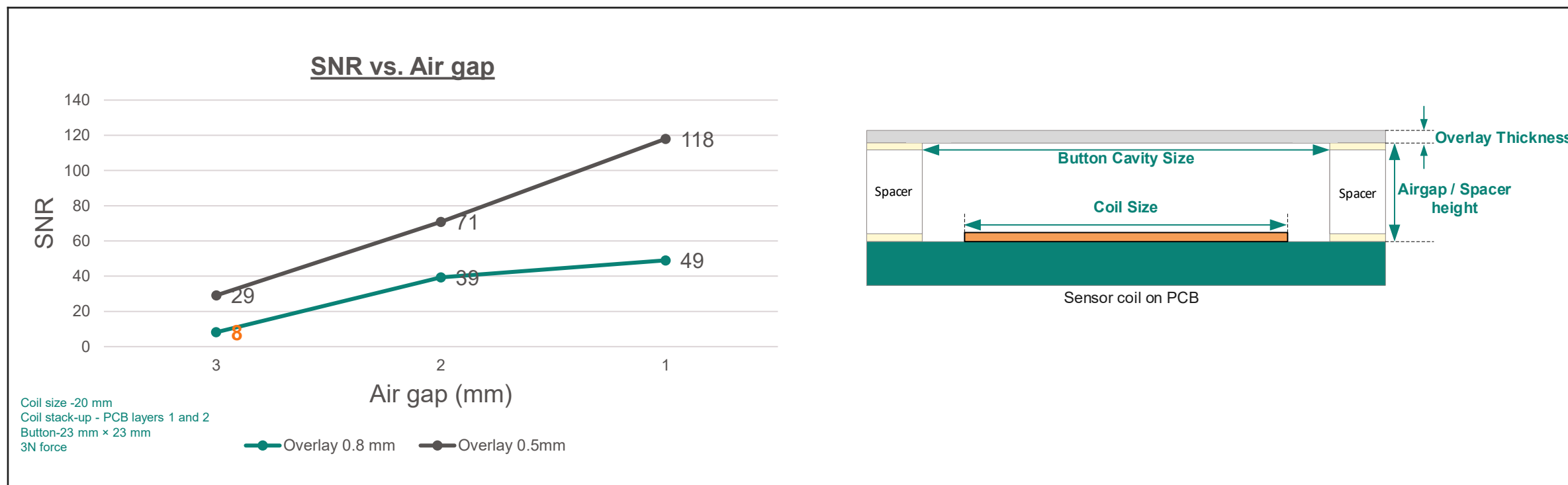
Button and coil size

- For optimum performance, the button cavity (overlay) size must be 133% of the coil/sensor size
- Larger button size means more deflection which results in better sensitivity



Airgap/spacer thickness

- As the airgap/spacing between the metal and the coil on the PCB is decreased, the sensitivity increases. This happens because the field lines coupling with the metal will be stronger closer to the PCB, which means the delta in inductance that is obtained is higher and easier to detect.

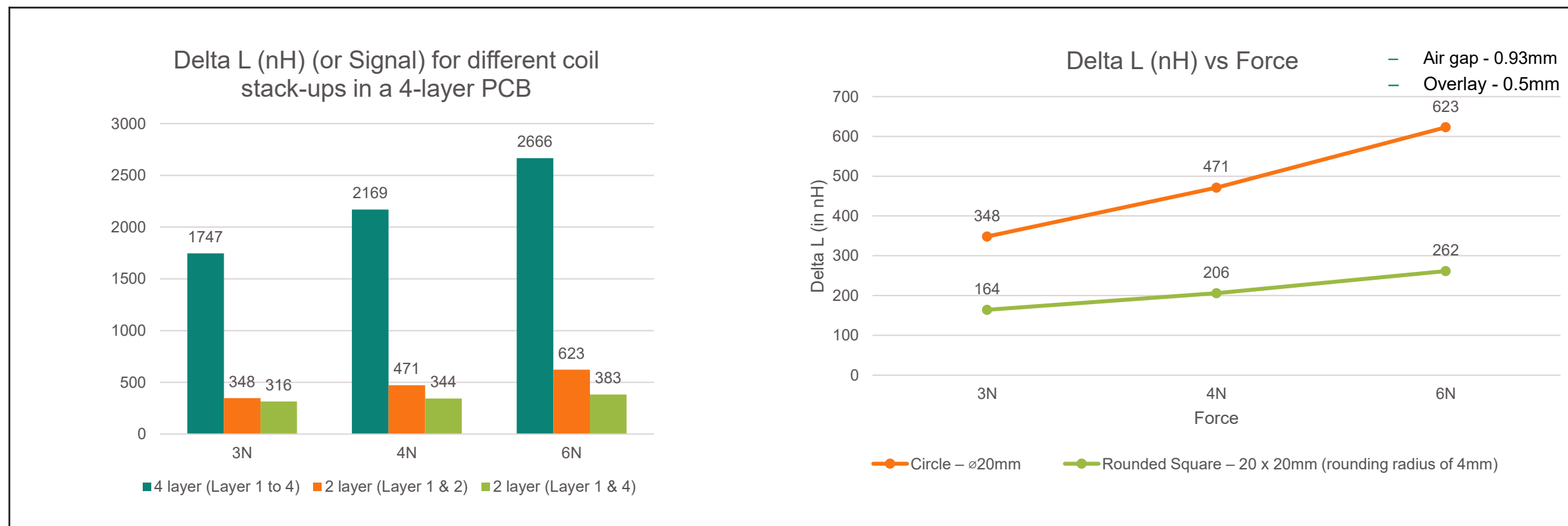


Overlay material

- The overlay material is selected based on the application requirements by analysing the mechanical, electrical, and magnetic properties and finding the right balance
- Materials such as stainless steel and aluminum can be used
 - Stainless steel is more durable and has a longer life span. It can withstand more force than aluminum before deforming permanently due to its higher yield strength. This makes it ideal for touch-over-metal (ToM) kind of applications
 - Alum has more magnetic conductivity compared to other commonly used metals, which makes it suitable for inductive sensing applications (ideal in case of proximity sensing/reed switch applications using Al as a metal target). It has a lower Young's modulus (elasticity) but can be permanently deformed

PCB specifications

- The more the number of layers the PCB is in (for the same coil diameter), the more the delta L (or signal) obtained, as shown by the graph
- Circular coils are recommended as they have higher SNR compared to rounded square coils



Overlay thickness

- The deflection of the metal used as an overlay depends on the material properties and metal thickness. Thinner metals are deflected more easily
- The more the deflection, the higher the change in the effective inductance and the higher the signal obtained. Therefore, the sensitivity drops with an increase in the overlay thickness. SNR can be increased by decreasing the metal thickness while keeping other parameters constant. However, the metal should not be so thin that it gets completely deformed when pressed.

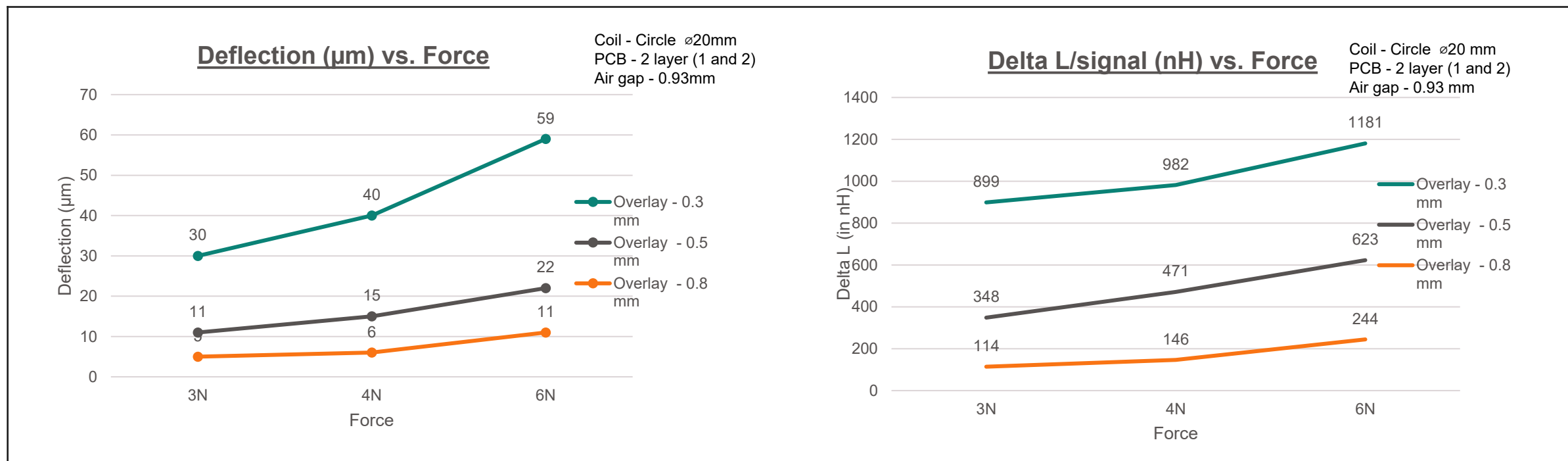
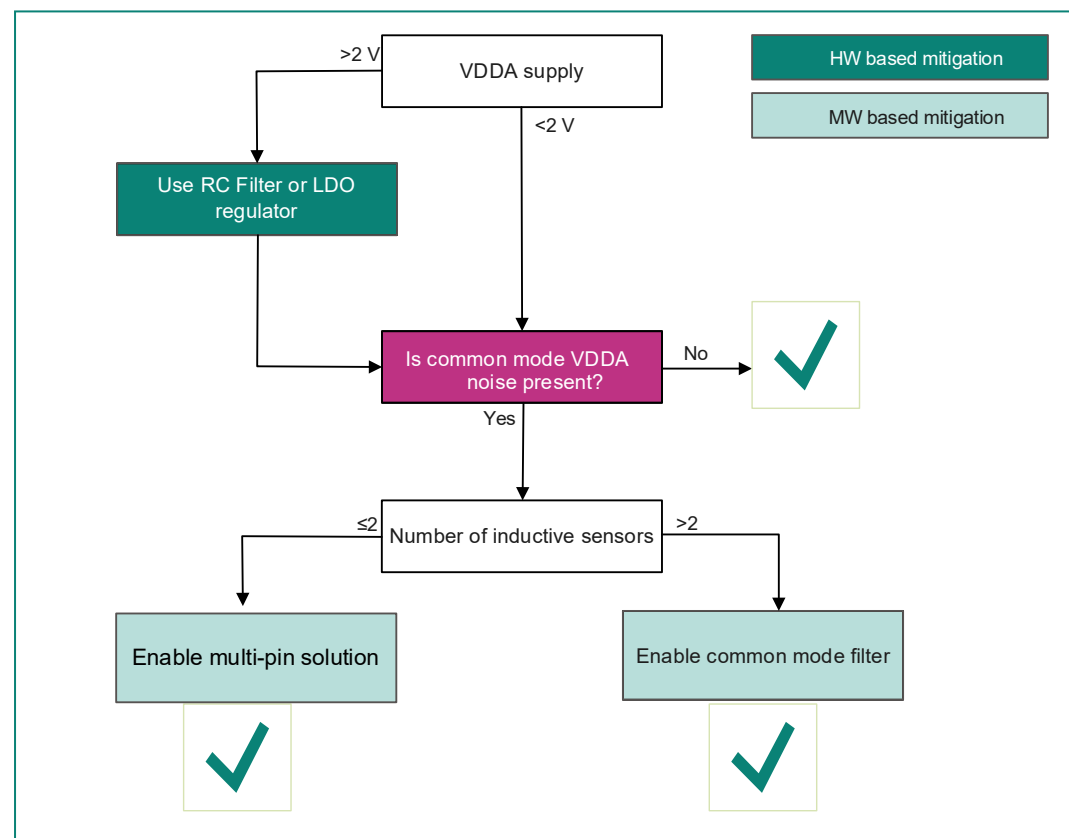
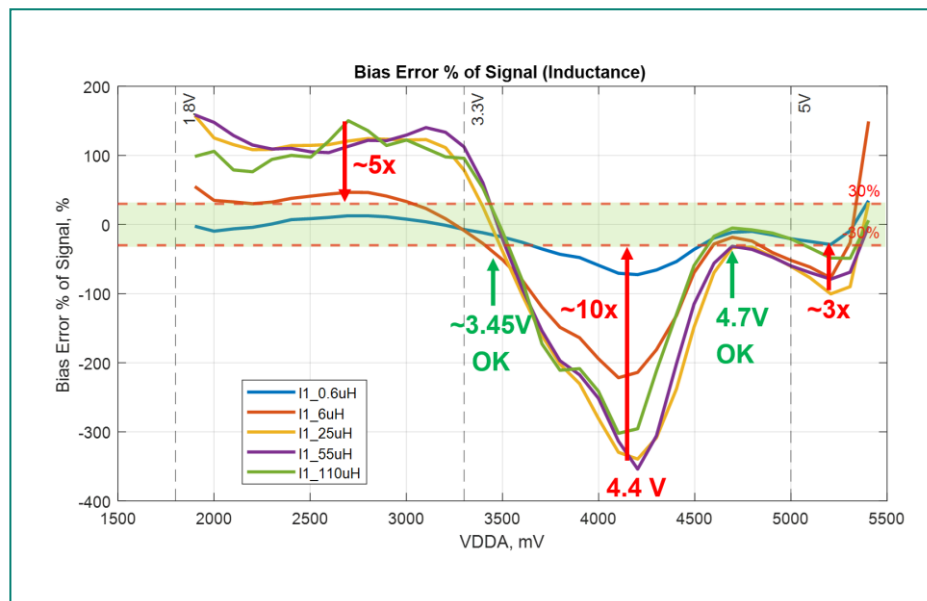


Table of contents

1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

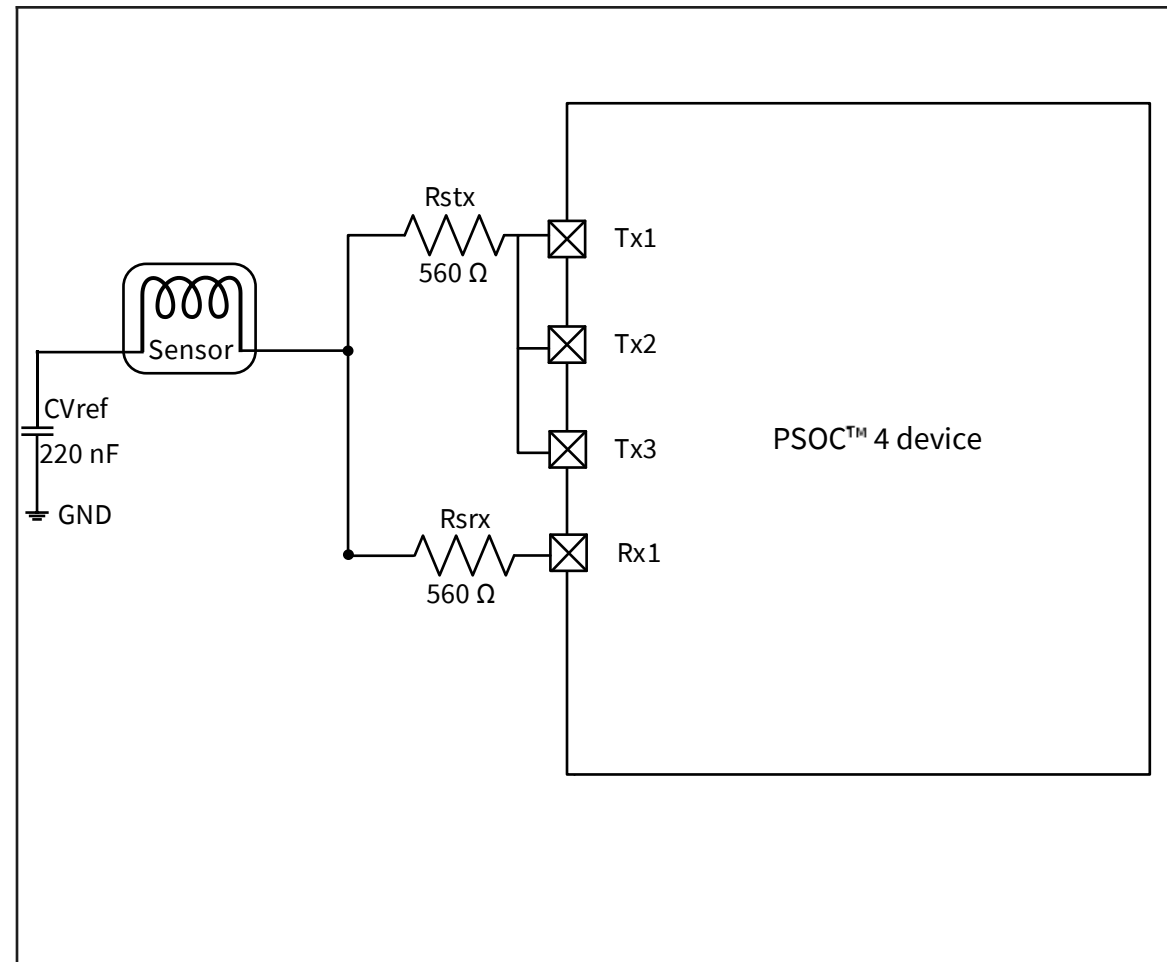
PST decision tree

- It is preferred to have an LDO regulator in the system to reduce the impact of power supply transients at all frequencies
- To understand each of the mitigation options in more detail, refer to the [ISX Design Guide](#)



PST mitigation options

- It is recommended to have an LDO regulator in the system to reduce the impact of power supply transients at all frequencies. Refer to the [ISX Design Guide](#) for suggested linear regulators
- Multi-pin configuration requires a hardware change and connection, as shown in the figure
- Common mode filter (CMF) is a FW based approach. The filter can only be applied when there are more than two sensors present in a widget



PST mitigation options (contd.)

Table 27 Mitigation options based on the PST noise frequency	
Frequency range of noise	Preferred mitigation option
0–30 Hz	CMF(Common mode filter) LDO (requires hardware) Multi-pin configuration (requires hardware configuration)
30 Hz – 1/Tscan	LDO (requires hardware) Multi-pin configuration (requires hardware configuration) (CMF is not recommended for this frequency range)
>1/Tscan	Less impact of power supply transients post the cut-off frequency.

where,

$$T_{scan} = \frac{LxClkDiv \times N_{sub}}{F_{mod}}$$

- Tscan = Scan time
- LxClkDiv = Lx clock divider
- Nsub= Number of sub-conversions
- Fmod = Modulation clock frequency (46 MHz by default)

Table of contents

1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

PSOC™ 4000T Multi-Sense Prototyping Kit (CY8CPROTO-040T-MS)

The CY8CPROTO-040T-MS PSOC™ 4000T Multi-Sense Prototyping Kit allows you to evaluate the features of the PSOC™ 4000T device.

The kit also comes with the following expansion boards:

- A touch-over-metal (ToM) Keypad-4 board with 14 mm coils
- A touch-over-metal (ToM) Keypad-2 board with 22 mm coils
- A flex PCB with 9 sensors, offering a full-scale liquid-level reading of 120 mm
- A hover-touch board with four 10 mm buttons that can detect a touch from a distance of 10 mm

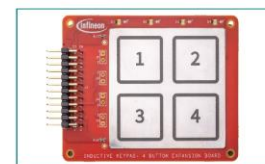
QUICK START GUIDE

PSOC™ 4000T Multi-Sense Prototyping Kit

CY8CPROTO-040T-MS

Kit contents

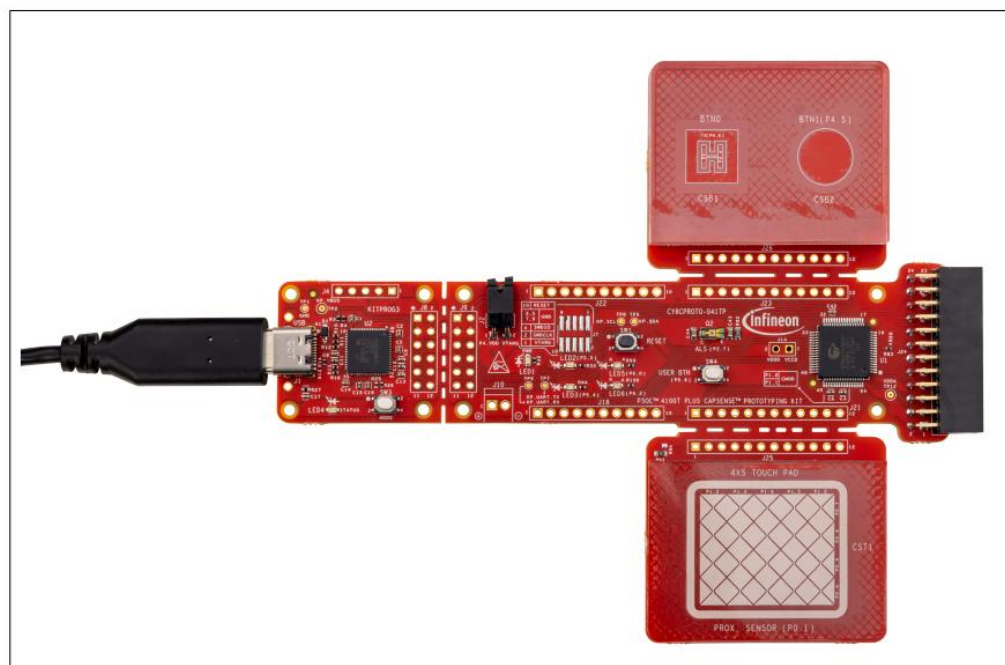
1. PSOC™ 4000T Multi-Sense control board
2. Inductive keypad - 4 expansion board
3. Inductive keypad - 2 expansion board
4. Liquid level sensor expansion board
5. Plastic container for liquid level sensing demonstration
6. CAPSENSE™ hover touch expansion board
7. USB-A to USB-C cable
8. Quick start guide (this document)



www.infineon.com/CY8CPROTO-040T-MS

CY8CPROTO-041TP PSoC™ 4100T Plus CAPSENSE™ Prototyping kit

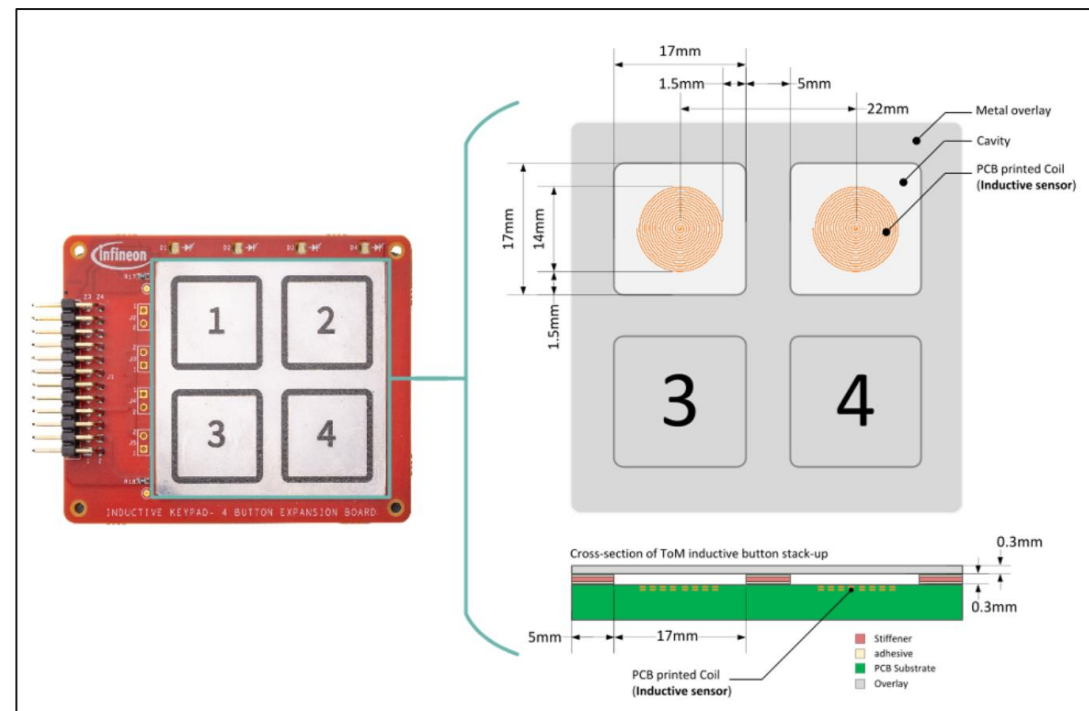
- The CY8CPROTO-041TP PSoC™ 4100T Plus CAPSENSE™ Prototyping kit enables evaluation and development of capacitive, inductive, and liquid-level sensing solutions, using the PSoC™ 4100T Plus device with fifth-generation CAPSENSE™ and multi-sense technology, offering an ultra-low-power touch HMI solution (MSCLP)
- The expansion boards that are part of the CY8CPROTO-040T-MS Multi-sense Prototyping Kit packaging are also compatible with this kit



Demo

Keypad-4 demo: SmartSense (beta version)

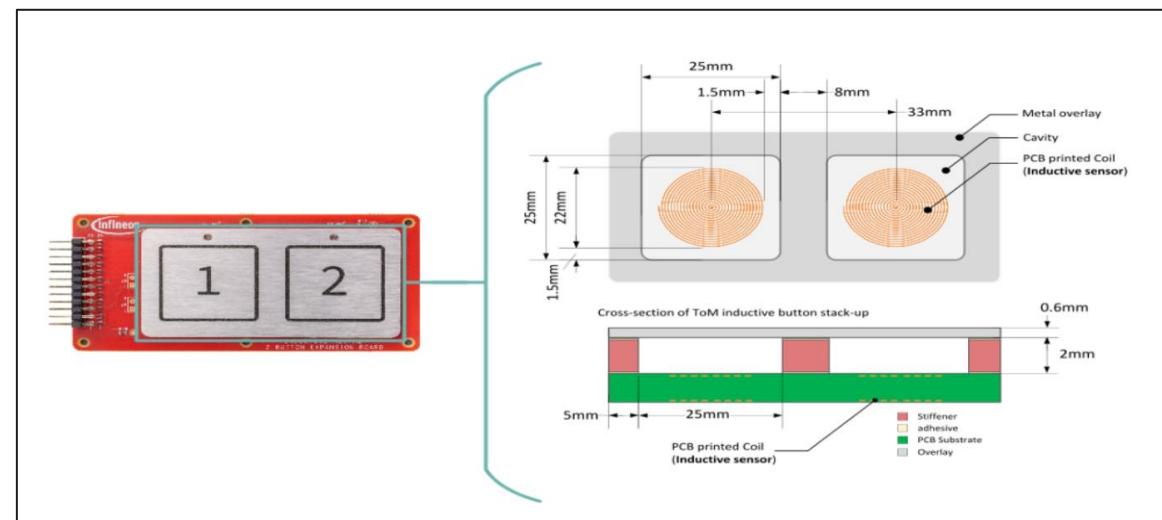
Parameter	Value
Number of sensors	4
Button shape and size	Rounded square, 17mm x 17 mm
Button center-to-center Spacing	22 mm
Sensor coil shape and size	Circle, 14 mm diameter
Stack-up of coil	2 layer (first 2 layers in a 4-layer board)
Metal overlay material	304-grade stainless steel
Metal overlay thickness	0.3 mm
Spacer or stiffener thickness	0.3 mm



Refer to the [code example](#) for more details.

Keypad-2

Parameter	Value
Number of sensors	2
Button shape and size	Rounded square, 25 mm x 25 mm
Button center-to-center spacing	33 mm
Sensor coil shape and size	Circle, 22 mm diameter
Stack-up of coil	2 layers (first two layers in a 2-layer board)
Metal overlay material	304-grade stainless steel
Metal overlay thickness	0.6 mm
Spacer or stiffener thickness	2 mm



Refer to the [code example](#) for more details.

Table of contents

1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

Manual tuning flow

The tuning procedure for the inductive sensors is explained in the following stages:

1. Set the initial parameters.
2. Measure the sensor inductance and set the sense clock frequency (LxClkDiv).
3. Set CDAC and dither parameters.
4. Fine-tune for the required SNR, power, and refresh rate.
5. Tune the threshold parameters.

Note: Refer to the [code example](#) for more details.

Table of contents

1	Introduction to Infineon's Inductive sensing (ISX)	3
2	Advantages of fly-back based inductive sensing	7
3	Sensor design and mechanical construction	13
4	Performance and reliability	21
5	CY8CPROTO-040T-MS kit details	25
6	Lab 1: Manual inductive sensor tuning process	30
7	Lab 2: SmartSense inductive sensor tuning process	32

SmartSense (beta version)

- ISX SmartSense helps you to get the basic tuning parameters and acts as a starting point for tuning a particular configuration
- SmartSense has the following two modes:
 - **Full auto-tune:** In this mode, the middleware sets both the hardware and the threshold parameters. This mode is not available for low-power widgets.
 - **Hardware parameters:** In this mode, the middleware sets the hardware parameters only and you must set the threshold parameters.
- To enable SmartSense, change the tuning method in the CAPSENSE™ configurator
- Touch sensitivity is the minimum sensitivity for which the configuration is tuned. This parameter can be changed to obtain the required SNR.
- [Keypad-4 code example](#) uses the SmartSense for tuning and delves in more details.

Revision history

Document revision	Date	Description of change
**	2025-12-11	Initial release.
*A	2026-04-07	Updated to align with Infineon branding guidelines.

